

Visual Symbols and the Blind

Can people who have never seen a picture understand what pictures mean? The answer is astonishing.

outlines • perspectives • literal representations • idiosyncratic • metaphorical • perimeter • consensus • ascribed • rendition • symbolism • raised-line drawings • motion lines • abstract shapes • visual metaphors • sighted

PART ONE • MOTION LINES AND THE SPINNING WHEEL

From a number of recent studies, it has become clear that blind people can appreciate the use of **outlines** and **perspectives** to describe the arrangement of objects and other surfaces in space. But pictures are more than **literal representations** — they also carry meaning through convention and symbol. This distinction raises a profound question: can someone who has never seen a picture understand what a picture means?

This fact was drawn to the author's attention dramatically when a blind woman in one of his investigations decided on her own **initiative** to draw a wheel as it was spinning. To show this motion, she traced a curve inside the circle. Lines of motion, such as the one she used, are a very recent invention in the history of illustration. Indeed, as art scholar David Kunzle notes, Wilhelm Busch, a trend-setting nineteenth-century cartoonist, used virtually no **motion lines** in his popular figures until about 1877.

When the author asked several other blind study subjects to draw a spinning wheel, one particularly clever **rendition** appeared repeatedly: several subjects showed the wheel's **spokes** as curved lines. When asked about these curves, they all described them as **metaphorical** ways of suggesting motion. Majority rule would argue that this device somehow indicated motion very well. But was it a better indicator than, say, broken or wavy lines — or any other kind of line, for that matter? The answer was not clear. So the author decided to test whether various lines of motion were apt ways of showing movement, or if they were merely **idiosyncratic** marks. Moreover, he wanted to discover whether there were differences in how the blind and the **sighted** interpreted **motion lines**.

"Can someone who has never seen a picture understand what a picture means? The evidence from blind subjects suggests the answer is a remarkable yes."

To search out these answers, the author created **raised-line drawings** of five different wheels, depicting **spokes** with lines that curved, bent, waved, dashed and extended beyond the **perimeter** of the wheel. He then asked eighteen blind **volunteers** to feel the wheels and assign one of the following motions to each wheel: wobbling, spinning fast, spinning steadily, jerking or braking. His control group consisted of eighteen **sighted** undergraduates from the University of Toronto.

All but one of the blind subjects assigned distinctive motions to each wheel. Most guessed that the curved **spokes** indicated that the wheel was spinning steadily; the wavy spokes, they thought, suggested that the wheel was wobbling; and the bent spokes were taken as a sign that the wheel was jerking. Subjects assumed that spokes extending beyond the wheel's **perimeter** signified that the wheel had its brakes on, and that dashed spokes indicated the wheel was spinning quickly.

18

blind volunteers tested
in the study

18

sighted undergraduates
in the control group

5

different raised-line
wheels used in the test

1877

year motion lines first
used in illustration

In addition, the favoured description for the **sighted** was the favoured description for the blind in every instance. What is more, the **consensus** among the sighted was barely higher than that among the blind. Because motion devices are unfamiliar to the blind, the task given to them involved some problem solving. Evidently, however, the blind not only figured out meanings for each line of motion, but as a group they generally came up with the same meaning at least as frequently as did sighted subjects.

◆ Key Finding — Part 1: Motion Lines

Blind **volunteers** interpreted **motion lines** in ways almost identical to **sighted** subjects, despite having no prior visual experience. This suggests that the conventions used in illustration are not arbitrary — they tap into something more fundamental about how the human mind interprets and represents movement and change, regardless of whether the person has ever seen a picture.

PART TWO · ABSTRACT SHAPES AND SYMBOLIC MEANING

The researchers found that the blind understand other kinds of **visual metaphors** as well. One blind woman drew a picture of a child inside a heart — choosing that symbol, she said, to show that love surrounded the child. With Chang Hong Liu, a **doctoral** student from China, the author began exploring how well blind people understand the **symbolism** behind shapes such as hearts that do not directly represent their meaning. This is a deeper question than the **motion lines** experiment: it asks whether blind people share the same emotional and cultural associations with **abstract shapes** as **sighted** people do.

The researchers gave a list of twenty pairs of words to **sighted** subjects and asked them to pick from each pair the term that best related to a circle and the term that best related to a square. For example: What goes with soft? A circle or a square? Which shape goes with hard?

All subjects deemed the circle soft and the square hard. A full 94% **ascribed** happy to the circle, instead of sad. But other pairs revealed less agreement: 79% matched fast to slow and weak to strong, respectively. And only 51% linked deep to circle and shallow to square. When the researchers tested four totally blind **volunteers** using the same list, they found that their choices closely resembled those made by the **sighted** subjects. One man, who had been blind since birth, scored extremely well — making only one match differing from the **consensus**, assigning 'far' to square and 'near' to circle. In fact, only a small majority of

sighted subjects — 53% — had paired far and near to the opposite partners. The researchers concluded that the blind interpret **abstract shapes** as sighted people do.

WORD PAIR	AGREEMENT AMONG SIGHTED SUBJECTS (%)
SOFT–HARD	100%
MOTHER–FATHER	94%
HAPPY–SAD	94%
GOOD–EVIL	89%
LOVE–HATE	89%
ALIVE–DEAD	87%
BRIGHT–DARK	87%
LIGHT–HEAVY	85%
WARM–COLD	81%
SUMMER–WINTER	81%
WEAK–STRONG	79%
FAST–SLOW	79%
CAT–DOG	74%
SPRING–FALL	74%
QUIET–LOUD	62%
WALKING–STANDING	62%
ODD–EVEN	57%
FAR–NEAR	53%
PLANT–ANIMAL	53%
DEEP–SHALLOW	51%

◆ Key Finding — Part 2: Abstract Shapes

Blind people interpret the symbolic meaning of **abstract shapes** — such as circles and squares — in virtually the same way as **sighted** people. This suggests that the associations we make between shapes and concepts such as softness, happiness, or warmth are not learned through vision, but are in some way universal — rooted in the human mind's way of organising experience, regardless of whether that experience has ever involved sight.

REFLECTION • WHAT DOES THIS TELL US?

The results of both experiments challenge a deeply held assumption: that visual symbols and conventions are the exclusive property of people who can see. If blind people can independently arrive at the same interpretations of **motion lines** and **abstract shapes** as **sighted** people — without ever having been taught

those conventions — then the symbolic language of pictures may be grounded in something far more fundamental than visual experience alone.

One explanation is that many visual symbols are **metaphorical** extensions of physical experience that all humans share — regardless of sight. A circle is soft and a square is hard perhaps because circles have no sharp edges and squares do. **Motion lines** curve and bend because real objects in motion curve and bend. If this is correct, then the conventions of illustration are not arbitrary cultural inventions but reflections of deeper patterns in how the human mind understands the world.

These findings also carry important practical implications. They suggest that images designed with blind people in mind — using **raised-line drawings**, tactile diagrams, and embossed illustrations — may be far more accessible and intuitive than previously assumed. The blind do not need to be taught a separate visual language. In many respects, they already speak it.

■ TALKING POINTS

Were you surprised that blind subjects interpreted motion lines in the same way as sighted subjects? Why / Why not?

The article suggests that some visual symbols are 'universal'. Can you think of any examples — or counterexamples?

What does this research tell us about the relationship between physical experience and symbolic thinking?

How might these findings change the way we design educational materials or public information for blind people?